

Learning With Noisy Labels for Industrial Time Series Outlier Detection: A Transformer-Embedded Contrastive Learning Framework

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Abstract—In many real-world industrial scenarios, acquiring accurately labeled data are often challenging due to limited resources or unexpected errors. Learning with noisy labels (LNL) has emerged as a significant research topic, aiming to develop reliable deep learning models using noisy-labeled training data. In this article, a novel Transformer-embedded LNL framework with fuzzy-clustering-assisted contrastive learning is developed for industrial time series outlier detection under noisy labels. Specifically, a fuzzy-clustering-assisted contrastive learning strategy is proposed to enhance the robustness of the Transformer encoder against noisy labels by leveraging the intrinsic characteristics of raw data. Furthermore, a dynamic two-stage training scheme is introduced to train the outlier detector. In the first training stage, the Transformer encoder is pretrained through data reconstruction to improve feature extraction capabilities for industrial time series. In the second stage, the outlier detector is jointly trained with the Transformer encoder, incorporating a joint learning strategy. Furthermore, a label-consistency regularization term is designed to enhance the robustness of the outlier detector against noisy labels by minimizing the discrepancy between the outputs of the outlier detector and the clustering algorithm. The proposed framework is applied to industrial time series data collected from a real-world wire arc additive manufacturing (WAAM) process. Experimental results demonstrate that the developed framework outperforms selected representative LNL approaches

in WAAM outlier detection under both low and high noise ratios.

Index Terms—Industrial time series analysis, learning with noisy labels (LNL), outlier detection, weakly supervised learning.

I. INTRODUCTION

DEEP learning (DL) has been extensively studied in recent years due to its remarkable ability to process large-scale data and its superior performance in feature extraction [17]. In DL, the quality of training data are crucial in determining the overall effectiveness of DL models. The accuracy of labels within the training dataset is of paramount importance, as mislabeled data (commonly referred to as noisy labels) may introduce misleading information during training, thereby impairing the model's generalization ability and degrading overall performance in real-world applications.

To mitigate the adverse effects of noisy labels on DL models, learning with noisy labels (LNL) has recently been investigated extensively, leading to the proposal of numerous approaches [9], [11], [15], [23], [33], [36]. Among existing LNL methods, sample selection, label correction, and robust training have been recognized as three prominent categories due to their satisfactory performance and ease of implementation. Specifically, sample selection and label correction approaches aim to enhance the quality of training labels by identifying correctly labeled samples and rectifying noisy labels, respectively. Meanwhile, robust training approaches focus on developing noise-robust training strategies or model architectures to mitigate the influence of noisy labels during model training.

It should be noted that sample selection approaches may lead to the exclusion of potentially valuable data, thereby reducing the diversity and completeness of the training set. Meanwhile, label correction approaches can introduce further errors if the corrected labels are inaccurate, ultimately misleading the training process and potentially reinforcing incorrect patterns in the model. Furthermore, each data point in an industrial time series contains essential intrinsic and structural information, which is critical for accurately identifying outliers, preserving temporal

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dependencies, and maintaining feature integrity in the presence of noisy labels [9].

To establish a reliable outlier detection model based on robust training approaches, the extraction of inherent patterns and features from industrial data is of critical importance. As a powerful family of feature extraction techniques, encoder-based methods have been widely employed in robust training-based LNL approaches due to their capability of capturing complex data representations [2], [25]. Among encoder-based methods, the Transformer has been particularly effective in handling time series data, primarily due to its self-attention mechanism [27]. The Transformer encoder allows multivariate time series to be processed simultaneously while capturing dependencies across different dimensions, thereby eliminating the need for data preprocessing [29].

It is worth mentioning that the standard training scheme of the Transformer encoder typically follows a supervised approach, which is susceptible to the adverse effects of noisy labels. In contrast, self-supervised learning (SSL) has emerged as a viable alternative, as it does not rely on label information during training. Among SSL techniques, contrastive learning has been particularly effective, as it aims to bring similar (i.e., positive) samples closer together while pushing dissimilar (i.e., negative) samples further apart. To date, contrastive learning has been widely adopted in robust training-based LNL approaches for constructing reliable encoders [14], [25].

Existing LNL approaches based on contrastive learning identify positive and negative sample pairs through data augmentation [13], [20]. Nevertheless, conventional data augmentation methods often fail to capture the intricate characteristics of time series data and tend to increase computational costs, particularly when handling large-scale datasets. To overcome these limitations, clustering algorithms, which group similar data points into distinct clusters based on inherent features, have been successfully employed for selecting positive and negative sample pairs in contrastive learning [4]. Recognized as a well-established clustering algorithm, the fuzzy C-means (FCM) algorithm enables data points to belong to multiple clusters with varying degrees of membership, leveraging fuzzy logic principles. Compared with hard clustering algorithms, FCM is not only easier to implement but also offers greater flexibility in handling complex data distributions. However, its performance may degrade when applied to high-dimensional data, primarily due to the presence of sparse and redundant features extracted by the Transformer encoder [8].

A seemingly natural solution for clustering high-dimensional data is to adopt the uniform manifold approximation and projection (UMAP) method, which serves as a competitive manifold-learning-based dimensionality reduction technique, in order to obtain a low-dimensional embedding from the high-dimensional input [1], [30]. The extracted low-dimensional representation can then serve as the input for the FCM algorithm, improving clustering effectiveness while mitigating the impact of irrelevant features. It is noticeable that clustering algorithms have been widely utilized in outlier detection as they are capable of uncovering the intrinsic structures and relationships within data,

thereby providing valuable insights. As such, beyond determining positive and negative sample pairs, the partitions generated by the FCM algorithm can also be leveraged for model regularization, further enhancing the model's generalization ability and overall performance.

Motivated by the above discussions, this article proposes a novel Transformer-embedded LNL framework with fuzzy-clustering-assisted contrastive learning (TFCCL) for outlier detection in industrial time series data with noisy labels. Specifically, a fuzzy-clustering-assisted contrastive learning (FCCL) approach is developed to train the Transformer encoder, where the UMAP-based FCM (UFCM) algorithm is introduced to determine positive and negative sample pairs for contrastive learning based on clustering results. A dynamic two-stage scheme is formulated for training the outlier detector. In the first training stage, the Transformer encoder is pretrained to enhance its feature extraction capability on industrial time series data through data reconstruction. In the second stage, the outlier detector is trained jointly with the Transformer encoder, while the UFCM algorithm is dynamically updated throughout the training process. Furthermore, a joint learning strategy is proposed for the second training stage, incorporating a label-consistency regularization term to minimize the discrepancy between outputs from the outlier detector and the UFCM algorithm, thereby improving the model's robustness against noisy labels.

The main contributions of this article are summarized as follows.

- 1) A novel FCCL strategy is proposed for training the Transformer encoder, where the UFCM algorithm is designed to process high-dimensional features and identify positive and negative sample pairs for contrastive learning.
- 2) A dynamic two-stage training scheme is introduced for the outlier detector, where the Transformer encoder is pretrained in the first stage via data reconstruction, and in the second stage, the outlier detector is jointly trained with the Transformer encoder.
- 3) A joint learning strategy is developed, incorporating a label-consistency regularization term to improve the model's generalization ability and robustness against noisy labels by minimizing the discrepancy between outputs from the outlier detector and the UFCM algorithm.
- 4) The proposed TFCCL framework is applied to outlier detection in real-world industrial time series data with noisy labels, specifically collected from a wire arc additive manufacturing (WAAM) pilot line. Experimental results validate the effectiveness of the proposed TFCCL framework.

The rest of this article is organized as follows. Section II discusses the background of LNL and WAAM while presenting the relevant preliminaries. In Section III, the proposed TFCCL framework and the training scheme are introduced. Experimental results for WAAM outlier detection under noisy labels are presented in Section IV. Finally, Section V concludes this article and provides discussions on potential future research directions.

II. BACKGROUNDS

A. Related Work

1) *Learning With Noisy Labels*: Existing LNL approaches can be broadly classified into two main categories: improving the quality of training data and mitigating the impact of noisy labels during the training process.

The first category of LNL approaches focuses on selecting training samples that are likely to have correct labels (i.e., clean samples) or on correcting noisy labels within the dataset. For instance, in [11], a classic approach known as coteaching has been proposed, where a Siamese network structure is employed, and each network selects low-loss samples to update its peer network, thereby improving training robustness. In [15], a sample selection approach called DivideMix has been developed, incorporating an improved semi-supervised training strategy that enables the model to learn from both selected clean samples and unselected samples. Furthermore, a joint training method with coregularization (JoCoR) has been introduced in [32], where a joint loss function with a coregularization term is designed to facilitate the selection of clean samples for training, thereby enhancing model stability in the presence of noisy labels.

The second category of LNL approaches focuses on mitigating the influence of noisy labels during the training process by designing robust DL models through specialized architectures or noise-resistant training strategies. These methods aim to enhance model robustness without directly modifying the training labels. For instance, in [25], a training framework has been proposed in which an encoder, combined with an SSL approach, is employed to improve the model's resistance to label noise. Similarly, in [26], a self-supervised adversarial noisy masking framework has been introduced, where adversarial noisy masking is applied to prevent the model from overfitting to noisy labels, while self-supervised reconstruction provides additional noise-free supervision, thereby improving model generalization in the presence of label noise.

2) *Outlier Detection in WAAM*: The WAAM is a type of additive manufacturing (AM) technology that utilizes a continuous metal wire feedstock and an electric arc as the heat source to melt the wire and deposit material layer by layer, thereby constructing physical objects. Compared with other AM technologies, WAAM is particularly suitable for manufacturing large-scale components and offers relatively higher deposition rates [6], [10]. Over the past few years, WAAM has been widely adopted across various industrial sectors, including aerospace, automotive, and power systems [10].

In WAAM, the electric arc serves as the primary heat source, and its operation is governed by the total current and arc voltage, which directly influence the melting process, deposition rate, and overall manufacturing quality. It should be noted that in real-world applications, sudden fluctuations in current and voltage may occasionally occur, potentially leading to defects such as incomplete fusion or uneven material deposition [12]. Over the past few years, outlier detection has attracted increasing attention in the research community due to its critical role in various real-world applications [18], [27], [31], [38]. As a result, in the field of WAAM, numerous approaches have been

developed to detect outliers, aiming to improve the WAAM process and ensure high-quality production [5], [12], [19]. For instance, in [5], a transfer learning approach has been introduced for outlier detection in WAAM data under different operational states. Furthermore, in [19], a semi-supervised method has been proposed to enable real-time anomaly detection in WAAM using current and voltage data, further enhancing the reliability and stability of the manufacturing process.

B. Preliminaries

In an M -class classification task involving noisy labels, consider a noisy dataset denoted as $\tilde{\mathcal{D}} = \{(x_i, \tilde{y}_i)\}_{i=1}^N$, where N denotes the number of samples, x_i is the i th sample, and $\tilde{y}_i$ is the corresponding observed (potentially noisy) label. Each sample in $\tilde{\mathcal{D}}$ is assumed to be independently and identically drawn from a joint distribution $P(X, \tilde{Y})$, which is defined over the space $\mathcal{X} \times \tilde{\mathcal{Y}}$. Here, $\mathcal{X} \subset \mathbb{R}^d$ represents the input feature space, while the label space is given by $\tilde{\mathcal{Y}} = \{\tilde{y} \in \{0, 1\}^M \mid \|\tilde{y}\|_1 = 1\}$, where each label $\tilde{y}$ is represented as a one-hot encoded vector, ensuring that only a single class label is assigned per sample.

III. METHODOLOGY

A. Motivation

In the presence of noisy labels, the supervised training process of the Transformer encoder may be adversely affected, leading to degraded feature extraction performance. To mitigate this issue, an effective approach is to reduce the model's reliance on ground-truth labels. As an SSL method, contrastive learning leverages the intrinsic relationships among samples to learn meaningful representations without requiring label supervision, making it a valuable technique for improving model robustness. In recent years, contrastive learning has been widely employed to assist model training, particularly in LNL. For instance, in [25], a self-supervised contrastive learning approach has been introduced for LNL to demonstrate its effectiveness in training both the encoder and classifier despite label noise.

It is worth noting that, in conventional contrastive learning, positive and negative sample pairs for each data point are typically identified using data augmentation methods. However, this approach is not well suited for large-scale time series datasets, as it may fail to capture complex temporal dependencies and could significantly increase computational costs. To overcome this limitation, a clustering-based approach for selecting positive and negative sample pairs presents a more efficient alternative. By leveraging the underlying structural information within the data, clustering enhances the correlation and diversity of positive and negative sample pairs while eliminating the need for additional label information, thereby improving the reliability of contrastive learning in noisy-label scenarios.

The FCM algorithm is a promising clustering technique for handling noisy and large-scale data, as it employs fuzzy logic to capture soft boundaries between clusters and assigns membership degrees to each data point across multiple clusters. Furthermore, the membership degrees generated by FCM can be used to regularize the classifier's output, thereby reducing

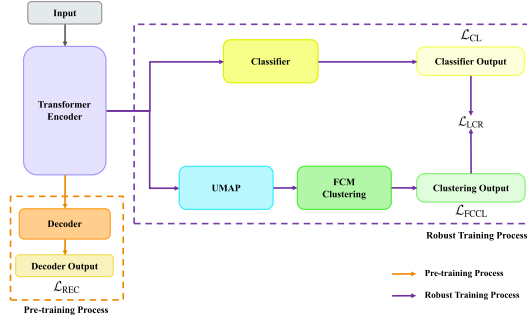


Fig. 1. Developed TFCCL framework, which consists of three key components: the Transformer encoder, the FCM clustering module, and the NN-based classifier.

the impact of label noise. To fully utilize both the inherent features and structural information in time series data for clustering, the Transformer encoder has recently been widely employed to extract effective feature representations [4]. However, directly using the high-dimensional features extracted by the Transformer encoder from large-scale time series datasets may degrade clustering performance. To overcome this challenge, applying dimensionality reduction techniques presents a natural and effective solution, as they can refine the extracted features by preserving essential structures while reducing computational complexity.

Motivated by the above discussions, in this article, a novel TFCCL framework is developed for outlier detection on time series data with noisy labels. Specifically, an FCCCL strategy is proposed to aid in training the Transformer encoder, where a UFCM algorithm is designed to process high-dimensional features and identify positive and negative sample pairs for contrastive learning. A dynamic two-stage scheme is introduced for training the outlier detector. In the first stage, the Transformer encoder is pretrained to enhance feature extraction capability on industrial time series data through data reconstruction. In the second stage, the outlier detector is jointly trained with the Transformer encoder, while the UFCM algorithm is dynamically updated throughout the training process. Furthermore, a joint learning strategy is proposed to improve the classifier's robustness against noisy labels, incorporating a label-consistency regularization term to minimize the discrepancy between the outputs of the outlier detector and the UFCM algorithm.

B. Overview of the TFCCL Framework

The developed TFCCL framework is illustrated in Fig. 1, which is designed to train a neural network-based (NN-based) classifier for time series outlier detection in the presence of noisy labels. The TFCCL framework consists of three key components: the Transformer encoder, the FCM clustering module, and the NN-based classifier. As shown in Fig. 2, the Transformer encoder architecture comprises multiple identical encoder layers, each consisting of a multihead attention mechanism and a convolutional module. Residual connections and layer normalization are applied to both the attention layer and the

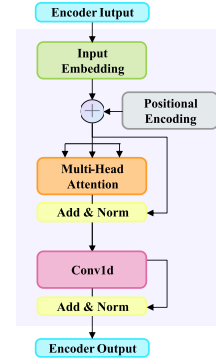


Fig. 2. Architecture of the Transformer encoder, which comprises multiple identical encoder layers. Each layer consists of a multihead attention mechanism and a convolutional module.

convolutional module to mitigate gradient vanishing. Furthermore, the NN-based classifier and the decoder are implemented as multilayer perceptrons. To enhance clustering performance, UMAP is employed for dimensionality reduction.

C. Loss Function

1) **Classification Loss:** The proposed TFCCL framework is designed to train a classifier for time series classification tasks. Although noisy labels may negatively impact supervised learning performance, incorporating the classification loss during training remains essential. In this article, cross-entropy is used to compute the classification loss. The classification loss $\mathcal{L}_{CL}$ is defined as

$$\mathcal{L}_{CL} = -\frac{1}{B} \sum_{i=1}^B \sum_{j=1}^M \tilde{y}_{ij} \log(\hat{y}_{ij}) \quad (1)$$

where B denotes the number of samples in the current mini-batch; M is the number of classes; $\tilde{y}_{ij}$ represents the j th component of the label for the i th sample; and $\hat{y}_{ij}$ is the j th component of the predicted output for the i th sample.

2) **Reconstruction Loss:** The reconstruction loss is employed for pre-training the Transformer encoder. Specifically, the mean squared error is used to calculate the reconstruction loss $\mathcal{L}_{REC}$, which is given by

$$\mathcal{L}_{REC} = \frac{1}{B} \sum_{i=1}^B \|x_i - \hat{x}_i\|^2 \quad (2)$$

where B represents the number of samples in the current mini-batch; and x_i and $\hat{x}_i$ are the true value and the predicted value of the i th sample, respectively.

3) **Contrastive Loss:** The contrastive loss is utilized to train the Transformer encoder. In this article, a novel fuzzy-clustering-assisted contrastive loss $\mathcal{L}_{FCC}$ is designed, which is shown as follows:

$$\mathcal{L}_{FCC} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\sum_{z_j \in \mathcal{P}(z_i)} \exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{z_k \in \mathcal{B}(z_i)} \exp(\text{sim}(z_i, z_k)/\tau)} \quad (3)$$

where B is the number of samples in the current mini-batch; $z_i \in \mathbb{R}^d$ denotes the feature representation of the i th sample; $\mathcal{P}(z_i)$ is the set of positive feature representations corresponding to the i th sample; $\mathcal{B}(z_i)$ is the set of all feature representations in the current mini-batch except the feature representation of the i th sample; $\text{sim}(\cdot, \cdot)$ is the cosine similarity function between two samples; and τ represents the temperature parameter, which is a positive constant and controls the sharpness of the similarity scores.

The contrastive loss in this article is formulated as a probability ratio that quantifies the likelihood of the model selecting the correct positive pair over all other sample pairs within the batch. The structure of the numerator and denominator reflects a probabilistic approach to pairwise discrimination, which emphasizes the relative similarity between positive and negative pairs. The exponential and logarithmic functions in (3) are combined in the contrastive loss to formulate a normalized, probabilistic objective based on the softmax and cross-entropy functions.

Remark 1: Traditional contrastive learning methods typically identify positive and negative sample pairs for the i th sample using augmented data or ground truth labels [20]. However, the large volume of data and the presence of noisy labels can interfere with the sample pairing process. To reduce computational costs and mitigate the effects of label noise, this article introduces the FCCL strategy, where positive and negative sample pairs for the i th sample are determined based on clustering results. Specifically, samples belonging to the same cluster as the i th sample are considered positive, while those in different clusters are treated as negative. The FCCL strategy enables the Transformer encoder to be trained in an unsupervised manner, thereby avoiding the adverse impact of noisy labels and improving its ability to extract meaningful feature representations.

4) Joint Learning Strategy: In this article, a label-consistency regularization term is introduced to quantify the discrepancy between the output membership of the UFCM algorithm and the output probability of the classifier. Specifically, the Kullback–Leibler (KL) divergence is employed to compute the label-consistency regularization term $\mathcal{L}_{\text{LCR}}$, which is defined as

$$\mathcal{L}_{\text{LCR}} = D_{\text{KL}}(P \parallel U) = \sum_{i=1}^M P(i) \log \left(\frac{P(i)}{U(i)} \right) \quad (4)$$

where M is the number of classes; P and U represent the distribution of the classifier's output probability and the distribution of the UFCM algorithm's output membership, respectively; and $P(i)$ and $U(i)$ represent the classifier's output probability for the i th class and the UFCM algorithm's output membership for the i th class, respectively.

Remark 2: The output membership of the UFCM algorithm is independent of the noisy labels in the training data, making it a valuable reference for mitigating the impact of label noise on the classifier. It is important to note that KL divergence is asymmetric, meaning that it quantifies the information loss incurred when one probability distribution is used to approximate another, with the divergence value varying based on the direction of comparison. Therefore, in this article, the KL divergence $D_{\text{KL}}(P \parallel U)$ is computed instead of $D_{\text{KL}}(U \parallel P)$ as

the label-consistency regularization term, aiming to better align the classifier's predictions with the UFCM algorithm's output membership.

5) Overall Loss Function: The overall loss function of the developed model consists of two parts: pretraining loss $\mathcal{L}_{\text{PRE}}$ and joint training loss $\mathcal{L}$. The pretraining loss $\mathcal{L}_{\text{PRE}}$ is given as follows:

$$\mathcal{L}_{\text{PRE}} = \mathcal{L}_{\text{REC}}. \quad (5)$$

The joint training loss $\mathcal{L}$ is given by

$$\mathcal{L} = \mu_1 \mathcal{L}_{\text{CL}} + \mu_2 \mathcal{L}_{\text{FCCL}} + \mu_3 \mathcal{L}_{\text{LCR}} \quad (6)$$

where μ_1 , μ_2 , and μ_3 are three hyperparameters to balance the weights among the classification loss, contrastive loss and label-consistency regularization term.

D. Dynamic Two-Stage Training Scheme

1) Pretraining of the Transformer Encoder in the First Stage:

The feature representations extracted from the training data by the Transformer encoder play a crucial role in the training process, as they are subsequently utilized by the classifier for classification and by the UFCM algorithm for contrastive learning and regularization. Although the FCCL strategy is employed to train the Transformer encoder, its performance is highly dependent on the quality of initialization. To ensure the effectiveness of FCCL and further enhance the Transformer encoder's feature extraction capabilities, this article employs a decoder to pretrain the encoder through input reconstruction. As illustrated in Fig. 1, during pretraining, only the encoder and decoder parameters are trainable, while all other parameters remain fixed. The reconstruction loss $\mathcal{L}_{\text{PRE}}$ is used as the loss function to guide the encoder and decoder in learning stable and robust feature representations. Pre-training allows the encoder to develop a deeper understanding of the underlying data structure, thereby improving training quality and overall model performance.

2) Dynamic Updating of the UFCM Algorithm in the Second Stage: In this article, UMAP is employed before applying the FCM algorithm to reduce the dimensionality of the Transformer encoder output. Furthermore, cosine similarity is used to measure the similarity between data points and cluster centroids, facilitating effective clustering of feature representations.

It should be noted that the updating frequency of the UFCM algorithm during training is a critical hyperparameter. A low updating frequency may cause the algorithm to become trapped in local optima, while a high updating frequency can lead to increased computational costs and unstable clustering performance. To address this issue, a dynamic updating strategy is introduced to adjust the UFCM algorithm's update frequency throughout the training process. Specifically, the update frequency is relatively high in the early training stages and gradually decreases as training progresses. The exact epoch for performing the $(k + 1)$ th UFCM update, denoted by e_{UFCM}^{k+1} , is calculated by

$$e_{\text{UFCM}}^{k+1} = \frac{k * (k + 1)}{2} * e_{\text{UFCM}}^k. \quad (7)$$

Algorithm 1: Training Procedure of the Developed TFCCL Framework.

Input: Noisy training data set $\tilde{\mathcal{D}} = \{(x_i, \tilde{y}_i)\}_{i=1}^N$, number of pre-training epochs E_P , number of training epochs E , the epoch number of the first UFCM update e_{UFCM}^1 , batch size B , hyper-parameters μ_1 , μ_2 and μ_3 .

Output: Model parameters θ .

```

1 Randomly initialize  $\theta$ .
2 Freeze  $\theta$  except the encoder and decoder.
3 for  $e \leftarrow 0$  to  $E_P$  do
4   | Update  $\theta$  of the encoder and decoder.
5 end
6 Unfreeze  $\theta$ .
7 Initialize UFCM using the encoder output.
8 for  $e \leftarrow 0$  to  $E$  do
9   for  $n \leftarrow 0, 1, \dots, \frac{|\tilde{\mathcal{D}}|}{B}$  do
10    | Randomly draw a mini-batch  $\{(x_i, \tilde{y}_i)\}_{i=1}^B$ .
11    | Calculate joint loss  $\mathcal{L}$  based on (6).
12    | Update  $\theta$ .
13  end
14  if  $e = e_{\text{UFCM}}^k$  then
15    | Update parameters of the UFCM algorithm.
16    | Update  $e_{\text{UFCM}}^k$  according to (7).
17  end
18 end

```

3) *Training Procedure:* The training procedure of the developed TFCCL framework is presented in Algorithm 1.

IV. OUTLIER DETECTION ON WAAM DATASETS WITH NOISY LABELS

A. Experimental Setup

1) *Data Description:* The data used in this article is collected from a pilot line for WAAM deployed in Sweden, which aims at developing an end-to-end digital solution by integrating automation methodologies for metal component manufacturing. Specifically, five datasets are employed in the experiments, each representing a time series corresponding to an individual manufacturing process. Each time series consists of 98 000 instances, where each instance contains measurements of welding current and voltage recorded at a sampling interval of 0.0002 s during the manufacturing process.

2) *Data Preprocessing:* In the experiments, each dataset is segmented using a sliding window approach, where the window length is set to 10 and the stride to 5. Each segment is assigned a label (“Normal” or “Outlier”), determined based on the labels of individual instances within the segment. Specifically, a segment is labeled as “Normal” if all instances within it are labeled as “Normal”; otherwise, it is classified as an “Outlier.” All labels are converted into a one-hot format for ease of classification. Each segmented dataset is then split into training and testing sets in a 7 : 3 ratio. Min-max normalization is subsequently applied

to both sets to scale the data to a uniform range, further improving the performance of outlier detection.

3) *Evaluation Metrics:* To comprehensively assess the performance of the developed TFCCL framework in WAAM outlier detection, essential evaluation metrics, such as accuracy, precision, recall, and F1-score are utilized. Each experiment is repeated five times, and the average values of these metrics are reported to ensure the consistency and reliability of the results.

4) *Baselines:* In this article, several novel and representative approaches are selected for comparison. In addition, a vanilla outlier detection method is employed as a standard baseline for comparison. The details of each selected approach are listed as follows.

- 1) *Coteaching* [11], where two deep neural networks (DNNs) are trained simultaneously and select low-loss samples to update each other.
- 2) *Coteaching* + [37], where the disagreement-based updating strategy is combined with the original coteaching framework.
- 3) *JoCoR* [32], where a coregularization term is employed to help select clean samples for model training.
- 4) *Colearning* [25], where supervised learning and self-supervised learning are combined to handle noisy labels.
- 5) *Standard Baseline*, where a classifier with the Transformer backbone is trained directly on the training data with the presence of noisy labels.

5) *Implementation Details:* For fair comparisons, all experiments are carried out on a server with NVIDIA RTX A6000 GPU with 48 GB memory and Intel(R) Xeon(R) Silver 4214R CPU. All approaches are implemented using Ubuntu 20.04.6, Pytorch 2.5.1, Python 3.9.21 and CUDA 12.3.

It is worth mentioning that convolutional neural networks (CNNs) are employed as the original backbone networks in selected LNL approaches. As such, in the experiments, both the CNN and the Transformer architectures are employed as the backbone networks for selected approaches for comparisons to fully investigate the competitive performance of the TFCCL framework.

To verify the effectiveness of the FCM algorithm in the TFCCL framework, the comparison experiment is conducted in this paper. Specifically, the K-means algorithm and the hierarchical clustering algorithm are chosen as baseline clustering algorithms to compare against the FCM used in the TFCCL framework. Besides, the dynamic updating strategy of the proposed UFCM algorithm is evaluated through comparative experiments. To be specific, linear and logarithmic strategies are selected as baselines for comparison.

The Transformer encoder consists of three encoder layers. The CNN backbone consists of five convolutional layers and a fully connected layer, where each convolutional layer is followed by batch normalization. The decoder and the classifier are a four-layer MLP and a three-layer MLP, respectively. In this article, the hyperparameters are selected via grid search on a held-out validation set. Each hyperparameter is explored from a range of values and the combination that achieves the best validation performance is selected. To be specific, the temperature parameter τ in contrastive loss $\mathcal{L}_{\text{FCCL}}$ is set to 0.08. The

TABLE I
OUTLIER DETECTION PERFORMANCE OF SELECTED APPROACHES ON WAAM DATASET 1 WITH DIFFERENT NOISE RATIOS

Approaches	Accuracy				Precision				Recall				F1 Score			
	30%	40%	50%	60%	30%	40%	50%	60%	30%	40%	50%	60%	30%	40%	50%	60%
Co-teaching (CNN)	99.03	96.21	75.35	71.76	98.54	93.35	49.45	49.24	95.95	93.74	77.76	70.21	97.22	91.94	56.75	53.05
Co-teaching+ (CNN)	99.00	96.54	78.79	75.78	98.28	93.79	52.85	53.58	95.53	93.09	80.50	75.06	97.30	93.17	60.49	57.60
JoCoR (CNN)	98.82	98.54	83.70	14.44	99.12	98.01	53.40	14.59	94.11	93.66	70.32	81.19	96.53	95.73	60.35	24.73
Co-learning (CNN)	98.36	96.78	80.0	78.29	98.3	94.08	54.24	56.47	95.42	93.34	82.37	77.20	97.48	94.02	67.16	63.48
Co-teaching (Transformer)	99.05	97.58	77.46	76.28	98.87	94.31	46.47	63.39	95.68	92.54	81.13	72.74	97.25	93.05	57.53	55.92
Co-teaching+ (Transformer)	99.43	97.8	83.06	82.55	99.05	94.55	53.16	60.42	95.57	92.97	85.27	80.50	97.33	93.35	66.82	64.63
JoCoR (Transformer)	97.96	97.67	80.54	62.76	99.37	99.49	72.04	82.99	95.03	94.11	87.17	70.82	97.15	96.73	76.49	74.64
Co-learning (Transformer)	99.33	97.95	86.24	83.14	98.87	94.45	54.78	60.51	95.75	93.08	86.83	82.67	97.39	93.60	68.33	65.96
Standard (Transformer)	90.22	66.73	44.68	10.37	86.64	67.98	60.48	18.72	91.02	89.89	36.82	12.36	92.84	77.41	45.77	14.89
TFCCCL	99.10	98.95	93.88	93.45	99.30	99.49	90.98	89.93	99.28	98.85	92.82	92.99	99.29	99.17	91.67	91.16

TABLE II
OUTLIER DETECTION PERFORMANCE OF SELECTED APPROACHES ON WAAM DATASET 2 WITH DIFFERENT NOISE RATIOS

Approaches	Accuracy				Precision				Recall				F1 Score			
	30%	40%	50%	60%	30%	40%	50%	60%	30%	40%	50%	60%	30%	40%	50%	60%
Co-teaching (CNN)	99.01	96.11	73.51	71.68	98.26	91.00	54.19	53.50	97.18	90.70	79.80	78.92	97.71	90.38	60.20	59.27
Co-teaching+ (CNN)	99.16	96.33	76.73	72.08	98.39	90.93	58.64	55.34	97.13	90.27	83.72	86.87	97.68	90.6	68.21	66.28
JoCoR (CNN)	98.95	98.94	80.21	15.89	97.56	97.42	50.43	16.18	97.17	97.27	94.04	79.69	97.36	97.34	65.41	26.88
Co-learning (CNN)	98.94	96.12	79.88	74.51	98.52	91.10	61.34	57.93	97.05	90.09	85.47	88.43	97.79	90.84	70.12	68.14
Co-teaching (Transformer)	99.17	96.92	76.48	68.53	98.34	94.31	60.60	61.36	97.46	90.11	84.65	83.08	97.89	92.05	65.02	64.73
Co-teaching+ (Transformer)	99.02	97.15	81.60	71.86	98.22	94.71	65.83	62.50	96.93	90.36	89.22	85.55	97.20	91.25	67.50	69.20
JoCoR (Transformer)	98.42	98.29	75.42	70.37	99.32	98.41	70.64	69.06	96.88	97.26	82.96	60.85	98.08	97.83	72.55	62.09
Co-learning (Transformer)	99.20	97.52	86.68	74.93	98.10	93.71	69.40	67.06	97.53	90.76	94.55	87.38	97.60	92.22	80.38	77.21
Standard (Transformer)	88.27	64.20	30.26	8.81	83.25	80.83	42.07	11.45	96.17	50.64	51.87	8.37	90.86	62.27	46.46	9.67
TFCCCL	98.84	98.66	91.19	89.02	98.98	98.83	91.60	92.18	99.28	98.85	92.82	92.99	99.01	98.38	89.23	86.09

TABLE III
OUTLIER DETECTION PERFORMANCE OF SELECTED APPROACHES ON WAAM DATASET 3 WITH DIFFERENT NOISE RATIOS

Approaches	Accuracy				Precision				Recall				F1 Score			
	30%	40%	50%	60%	30%	40%	50%	60%	30%	40%	50%	60%	30%	40%	50%	60%
Co-teaching (CNN)	99.05	96.27	76.62	71.31	98.05	89.72	54.44	48.97	96.94	91.26	71.08	67.39	97.47	90.36	56.16	51.64
Co-teaching+ (CNN)	98.90	96.57	80.51	78.21	98.17	89.47	58.13	53.41	96.84	91.66	77.69	75.88	97.50	90.55	66.48	63.58
JoCoR (CNN)	99.14	98.99	81.32	15.16	98.40	98.65	50.54	15.09	97.02	96.00	94.09	77.91	97.70	97.30	65.57	25.27
Co-learning (CNN)	98.78	96.87	84.23	81.06	98.02	89.92	62.44	56.53	96.96	91.36	81.23	78.66	97.43	90.63	70.65	65.75
Co-teaching (Transformer)	99.13	96.56	85.88	82.39	98.64	91.51	61.73	53.58	96.74	90.31	72.93	62.38	97.67	90.86	65.55	57.55
Co-teaching+ (Transformer)	98.98	96.86	86.08	85.69	98.54	91.91	57.23	56.68	96.12	90.01	77.53	65.88	96.84	90.78	67.03	59.58
JoCoR (Transformer)	98.49	98.25	79.27	63.32	99.25	98.90	70.19	74.33	96.61	95.73	86.75	78.55	97.91	97.29	75.52	66.41
Co-learning (Transformer)	98.83	96.56	87.28	85.39	98.36	92.16	71.73	59.88	96.82	89.46	82.33	70.28	97.73	90.60	73.53	61.58
Standard (Transformer)	77.04	58.59	35.88	9.01	72.47	61.71	47.15	8.88	93.98	82.92	50.58	5.46	84.04	70.76	48.80	6.76
TFCCCL	98.91	98.78	93.71	91.95	99.55	98.23	94.35	92.24	98.65	98.40	90.16	87.50	99.10	98.31	91.53	89.45

hyperparameters μ_1 , μ_2 , and μ_3 in the training loss $\mathcal{L}$ are set as 0.3, 0.8, and 0.8, respectively. In the experiments, the Adam optimizer is utilized for model training, where the learning rate is 0.0001. The pre-training and training epochs are 15 and 50, respectively. During the training process, the epoch number for the first UFCM update e_{UFCM}^1 is set to 3.

The noise ratio λ is an important parameter in LNL, which indicates the ratio of the noisy labels in a dataset. It should be noted that the outlier detection task in this article is formulated as a binary classification problem with only two classes. As such, symmetrical label noise is employed in the experiment, where the label of each data point is corrupted with an equal probability defined by λ . For a comprehensive evaluation, multiple experiments are conducted on each dataset with various values

of λ (e.g., 30%, 40%, 50%, and 60%). It should be noted that in real-world industrial scenarios, the noise ratio usually remains below 50%. To verify the outlier detection performance of the TFCCCL framework under severe label noise, the WAAM data sets with a noise ratio of 60% are utilized in the experiments.

B. Evaluation of WAAM Outlier Detection

The results of WAAM outlier detection on five datasets with noisy labels using selected approaches and the TFCCCL framework are illustrated in Tables I–V. According to the results, the TFCCCL framework shows the leading performance on five datasets with five different noise ratios. The selected approaches using the Transformer backbone demonstrate better

TABLE IV
OUTLIER DETECTION PERFORMANCE OF SELECTED APPROACHES ON WAAM DATASET 4 WITH DIFFERENT NOISE RATIOS

Approaches	Accuracy				Precision				Recall				F1 Score		
	30%	40%	50%	60%	30%	40%	50%	60%	30%	40%	50%	60%	30%	40%	60%
Co-teaching (CNN)	96.17	90.82	61.97	60.01	81.41	66.76	35.72	31.74	98.07	83.91	75.61	72.90	88.86	73.58	44.60
Co-teaching+ (CNN)	96.32	90.52	67.02	62.33	81.29	67.16	38.47	37.80	97.92	84.21	80.11	75.93	88.89	74.71	50.17
JoCoR (CNN)	98.27	96.82	73.63	12.18	91.18	83.70	37.76	11.79	98.40	99.08	99.89	74.72	94.64	90.66	20.37
Co-learning (CNN)	96.20	90.82	70.97	65.01	81.17	67.41	43.22	39.54	98.03	83.91	84.31	78.40	88.71	75.01	56.92
Co-teaching (Transformer)	96.54	90.77	66.33	63.53	83.78	65.71	37.89	30.87	96.68	86.47	83.96	79.48	89.68	74.35	49.32
Co-teaching+ (Transformer)	96.39	91.07	71.83	67.03	83.96	65.31	42.09	33.87	96.56	86.97	88.76	81.98	89.82	74.60	47.94
JoCoR (Transformer)	96.41	96.55	71.65	62.72	96.42	94.09	61.44	53.52	94.91	96.85	84.22	96.12	95.64	95.40	68.03
Co-learning (Transformer)	96.20	91.37	77.03	71.13	83.80	64.91	47.39	37.57	96.45	87.37	93.56	85.38	89.65	74.90	62.70
Standard (Transformer)	74.32	61.58	15.75	6.09	71.92	64.56	31.52	15.11	75.78	41.60	23.94	9.26	83.67	58.75	27.21
TFCCCL	99.03	98.45	95.48	93.98	99.02	98.57	93.14	89.62	99.51	98.00	94.15	94.37	99.26	98.27	93.47

TABLE V
OUTLIER DETECTION PERFORMANCE OF SELECTED APPROACHES ON WAAM DATASET 5 WITH DIFFERENT NOISE RATIOS

Approaches	Accuracy				Precision				Recall				F1 Score		
	30%	40%	50%	60%	30%	40%	50%	60%	30%	40%	50%	60%	30%	40%	60%
Co-teaching (CNN)	98.80	96.56	73.93	68.36	96.91	89.33	44.45	44.42	95.36	89.94	69.53	69.04	96.10	89.15	49.43
Co-teaching+ (CNN)	98.92	96.80	77.62	75.12	96.74	89.80	49.00	46.67	95.51	89.59	74.63	72.59	95.70	89.66	55.73
JoCoR (CNN)	99.04	98.78	82.40	12.54	96.06	96.75	47.45	12.21	97.86	95.37	95.59	77.20	96.93	96.04	21.08
Co-learning (CNN)	96.17	90.87	71.09	65.11	81.41	66.86	43.17	39.29	97.74	84.61	80.89	79.20	88.90	74.68	57.01
Co-teaching (Transformer)	98.83	97.21	79.10	76.99	98.76	92.46	44.61	39.89	97.65	89.97	69.90	70.27	98.20	90.94	52.18
Co-teaching+ (Transformer)	98.68	97.56	84.33	80.13	98.64	92.91	50.28	42.27	97.81	89.72	74.79	73.60	98.12	90.64	53.25
JoCoR (Transformer)	98.14	97.85	78.64	64.41	99.39	98.36	69.62	64.99	94.92	95.72	87.96	83.00	97.10	97.02	72.97
Co-learning (Transformer)	98.53	97.86	86.53	84.23	98.52	92.61	55.03	45.72	97.26	90.12	79.29	77.80	97.63	90.39	66.48
Standard (Transformer)	79.42	73.21	47.45	7.07	84.60	83.31	69.09	14.23	84.81	75.20	39.43	7.62	84.71	79.05	50.21
TFCCCL	98.95	98.42	95.20	92.22	99.42	98.06	90.10	87.58	99.01	99.62	96.51	89.53	99.21	98.83	93.07

performance than those using the CNN backbone. A detailed evaluation of different noise ratios is presented below.

1) 30% and 40% Noisy Labels: It can be found that all selected approaches with different backbones as well as the TFCCCL framework all achieve satisfactory performance on five data sets. The standard approach also shows acceptable performance under 30% noise ratio. As indicated by the results, the TFCCCL framework outperforms the other approaches on most datasets in terms of accuracy, precision, recall, and F1 score. Notably, on dataset 2 with a 30% noise ratio, coteaching, coteaching+ and colearning (with both backbones) all achieve higher accuracy than TFCCCL. Datasets 1 and 2 with noise ratios of 30%, colearning-transformer has higher accuracy than TFCCCL. On datasets 2, 3, and 5 with noise ratios of 30% and 40%, JoCoR-CNN has higher accuracy than TFCCCL as well. Nevertheless, TFCCCL attains the highest performance on the remaining metrics in these scenarios, which demonstrates the overall effectiveness of TFCCCL in outlier detection with a relatively small number of labels.

2) 50% Noisy Labels: In the scenario with a 50% noise ratio, the TFCCCL remains superior performance, whereas the performance of all selected approaches decreases significantly on all five datasets. Specifically, the accuracy and the F1 score of the selected approaches are no more than 88% and 78%, respectively. For TFCCCL, the accuracy and the F1 score across five datasets exceed 91% and 88%, respectively, which shows the robustness of TFCCCL against a moderate number of noisy labels.

3) 60% Noisy Labels: In situations with the noise ratio of 60%, the baseline approach and JoCoR-CNN are affected by severe label noise and are no longer capable of handling outlier detection tasks. The performance of the other selected approaches is also not effective enough. It should be noted that TFCCCL still achieves satisfactory performance across five data sets, with each dataset's accuracy above 89% and F1 score above 86%. The results indicate the competitive performance of the TFCCCL in outlier detection with a large number of noisy labels.

C. Comparison Experiments

In this article, two comparison experiments are conducted. Specifically, the first experiment aims to verify the effectiveness of the selection of the FCM algorithm. The results are illustrated in Table VI. According to the results, the FCM algorithm shows competitive performance across all five datasets under different noise ratios compared with other hard clustering algorithms, demonstrating its robustness and adaptability in the presence of label noise. The second experiment aims to verify the effectiveness of the proposed dynamic updating strategy for the UFCM algorithm. As shown in Table VII, the developed dynamic updating strategy achieves the best results compared with other updating strategies.

D. Ablation Study

In this article, a detailed ablation study is carried out to verify the effectiveness of the main components in TFCCCL

TABLE VI

COMPARISON OF DIFFERENT CLUSTERING ALGORITHMS WITHIN THE TFCCL FRAMEWORK ON WAAM DATASETS UNDER VARYING NOISE RATIOS

Approaches	Metrics (%)	Data Set 1				Data Set 2				Data Set 3				Data Set 4				Data Set 5			
		30%	40%	50%	60%	30%	40%	50%	60%	30%	40%	50%	60%	30%	40%	50%	60%	30%	40%	50%	60%
TFCCL-Kmeans	Accuracy	98.12	96.22	89.02	84.23	97.84	95.68	86.43	80.48	97.97	96.09	88.13	82.47	98.07	95.28	89.52	83.08	97.91	95.42	89.03	82.91
	Precision	98.29	96.42	85.08	80.11	97.72	95.01	85.59	78.52	98.28	95.01	88.31	80.88	97.81	95.11	86.48	78.59	98.12	94.48	83.28	74.01
	Recall	98.34	96.09	86.12	77.68	97.81	94.01	79.02	70.01	96.68	94.32	82.52	74.48	98.42	94.02	86.12	79.79	97.62	96.58	87.92	71.01
	F1 Score	98.30	96.26	85.59	78.84	97.76	94.48	82.12	73.83	97.48	94.62	85.31	77.41	98.11	94.53	86.29	79.26	97.84	95.52	85.48	72.31
TFCCL-Hierarchical	Accuracy	95.58	90.97	82.33	75.47	94.83	90.28	80.24	72.36	95.13	90.26	81.16	73.84	95.18	89.62	80.76	72.28	94.68	89.78	80.08	71.96
	Precision	95.76	91.04	78.86	72.84	94.42	89.27	77.47	69.87	95.33	89.32	80.07	71.94	94.57	88.84	77.94	68.57	94.64	87.83	75.08	65.96
	Recall	95.27	90.23	78.16	70.48	94.34	87.77	70.77	62.27	94.02	87.73	74.16	65.47	95.04	86.28	75.36	66.07	93.78	88.03	79.06	63.44
	F1 Score	95.51	90.58	78.48	71.57	94.37	88.53	73.97	65.58	94.61	88.55	76.97	68.63	94.76	87.57	76.58	67.18	94.18	87.46	76.87	64.52
TFCCL-FCM	Accuracy	99.10	98.95	93.88	93.45	98.84	98.66	91.19	89.02	98.91	98.78	93.71	91.95	99.03	98.45	95.48	93.98	98.95	98.42	95.20	92.22
	Precision	99.30	99.49	90.98	89.93	98.98	98.83	91.60	92.18	98.55	98.23	94.35	92.24	99.02	98.57	93.14	89.62	99.42	98.06	90.10	87.58
	Recall	99.28	98.85	92.82	92.99	99.04	97.94	87.65	82.10	98.65	98.40	90.16	87.50	99.51	98.00	94.15	94.37	99.01	99.62	96.51	89.53
	F1 Score	99.29	99.17	91.67	91.16	99.01	98.38	89.23	86.09	99.10	98.31	91.53	89.45	99.26	98.27	93.47	91.48	99.21	98.83	93.07	88.37

TABLE VII

COMPARISON OF DIFFERENT UFCM UPDATING STRATEGIES WITHIN THE TFCCL FRAMEWORK ON WAAM DATASETS UNDER VARYING NOISE RATIOS

Approaches	Metrics (%)	Data Set 1				Data Set 2				Data Set 3				Data Set 4				Data Set 5			
		30%	40%	50%	60%	30%	40%	50%	60%	30%	40%	50%	60%	30%	40%	50%	60%	30%	40%	50%	60%
TFCCL (Linear)	Accuracy	98.26	94.87	88.65	73.00	95.70	91.78	85.43	76.62	95.57	89.94	85.27	84.36	92.80	90.37	83.99	77.69	95.80	92.76	86.34	80.14
	Precision	97.78	92.50	90.02	71.98	98.23	91.67	82.73	78.79	94.07	93.50	84.98	88.31	92.71	94.08	89.13	90.84	99.34	98.81	77.78	87.80
	Recall	99.54	97.76	89.18	97.62	92.63	94.53	96.52	84.42	98.90	89.57	93.82	85.09	96.66	91.10	86.47	75.60	93.79	90.32	96.23	86.91
	F1 Score	98.67	96.10	89.97	83.27	96.17	93.07	89.58	82.02	96.44	91.50	88.81	86.40	94.66	92.57	88.22	87.25	96.77	94.36	88.09	87.64
TFCCL (Logarithmic)	Accuracy	96.93	92.56	86.20	70.64	94.47	89.13	83.14	74.54	94.06	87.58	81.70	82.28	90.94	88.17	81.00	75.41	94.15	90.48	78.07	77.53
	Precision	96.65	90.00	87.83	70.42	96.78	88.81	80.03	76.95	92.54	91.31	82.13	86.46	91.40	91.72	86.79	88.57	97.55	96.57	75.33	85.58
	Recall	98.03	95.67	86.28	95.30	90.78	92.13	93.64	82.83	97.55	87.41	91.35	83.37	95.62	88.37	84.15	73.93	92.57	87.93	94.20	85.29
	F1 Score	97.10	93.21	87.81	81.36	94.92	90.50	86.69	80.03	94.82	89.16	86.19	84.83	93.50	89.77	85.82	85.32	94.95	91.70	86.62	85.53
TFCCL (Dynamic)	Accuracy	99.10	98.95	93.88	93.45	98.84	98.66	91.19	89.02	98.91	98.78	93.71	91.95	99.03	98.45	95.48	93.98	98.95	98.42	95.20	92.22
	Precision	99.30	99.49	90.98	89.93	98.98	98.83	91.60	92.18	98.55	98.23	94.35	92.24	99.02	98.57	93.14	89.62	99.42	98.06	90.10	87.58
	Recall	99.28	98.85	92.82	92.99	99.04	97.94	87.65	82.10	98.65	98.40	90.16	87.50	99.51	98.00	94.15	94.37	99.01	99.62	96.51	89.53
	F1 Score	99.29	99.17	91.67	91.16	99.01	98.38	89.23	86.09	99.10	98.31	91.53	89.45	99.26	98.27	93.47	91.48	99.21	98.83	93.07	88.37

TABLE VIII

RESULTS OF ABLATION STUDY ON WAAM DATASETS WITH DIFFERENT NOISE RATIOS

	Metrics (%)	Data Set 1				Data Set 2				Data Set 3				Data Set 4				Data Set 5			
		30%	40%	50%	60%	30%	40%	50%	60%	30%	40%	50%	60%	30%	40%	50%	60%	30%	40%	50%	60%
w/o Pre-training	Accuracy	95.17	93.96	80.26	71.22	95.75	94.02	77.57	63.44	96.75	94.52	86.50	76.94	95.48	94.93	86.85	80.80	95.70	93.98	88.49	82.74
	Precision	99.84	96.42	82.23	70.66	95.77	92.07	75.03	61.49	94.95	91.71	90.85	75.34	93.65	92.85	89.35	83.48	95.84	92.63	92.85	96.35
	Recall	92.38	93.97	87.85	93.43	97.00	98.19	92.24	99.85	99.94	99.97	96.35	91.92	99.90	99.87	90.85	88.29	97.85	98.91	89.77	74.31
	F1 Score	96.04	95.18	84.95	80.46	96.38	95.03	82.75	76.11	97.38	95.66	88.54	82.81	96.67	96.29	90.09	85.81	96.83	95.67	91.29	85.26
w/o FCCL	Accuracy	92.74	92.18	46.50	11.99	94.97	94.01	72.39	27.62	95.29	93.06	66.07	34.54	88.13	85.02	52.50	46.65	94.00	93.16	41.72	20.68
	Precision	95.46	97.20	55.76	24.68	93.48	93.43	61.42	34.59	97.34	96.21	80.14	41.76	84.71	81.45	65.41	39.04	94.29	95.44	51.29	38.66
	Recall	91.95	90.27	50.66	18.91	98.22	96.53	59.10	27.03	94.79	92.15	56.25	21.11	98.04	93.54	61.84	18.90	96.94	94.33	56.15	30.78
	F1 Score	93.68	93.60	52.98	21.41	95.79	94.95	64.10	30.34	96.05	94.13	65.42	28.04	91.72	89.78	63.53	31.79	95.59	94.88	52.80	34.27
w/o UMAP	Accuracy	98.85	95.61	64.85	57.82	96.79	93.59	69.91	56.75	96.29	88.66	76.97	64.64	93.47	90.97	67.91	65.31	96.36	93.40	65.49	59.80
	Precision	99.04	93.90	66.89	69.83	96.53	94.21	70.33	62.12	97.60	92.88	76.06	65.28	90.97	98.63	70.38	59.61	95.94	95.95	71.53	71.87
	Recall	97.80	99.54	88.26	58.97	98.02	94.84	83.76	66.27	96.23	84.19	84.01	88.63	96.45	87.49	88.42	69.73	98.76	94.15	80.81	66.01
	F1 Score	98.41	96.64	76.08	63.94	97.27	94.52	76.46	64.13	96.91	91.42	82.54	75.18	95.27	92.73	78.38	64.79	97.33	95.04	75.89	68.81
w/o Dynamic UFCM Updating Strategy	Accuracy	98.28	94.83	84.76	70.83	95.70	91.75	84.32	75.20	95.58	89.90	82.62	81.92	92.81	90.34	82.13	76.29	95.78	92.72	79.05	78.57
	Precision	97.79	92.48	88.03	69.64	98.22	91.63	81.46	76.59	94.08	93.48	83.68	86.47	92.71	94.05	87.39	88.82	99.95	98.78	76.23	83.53
	Recall	99.54	99.97	87.93	95.76	92.62	94.49	94.66	82.80	98.90	89.53	92.19	83.08	96.66	91.08	85.11	73.51	93.77	90.28	94.31	84.84
	F1 Score	98.66	96.08	87.98	80.64	96.17	93.04	87.57	79.57	96.43	91.46	87.28	84.74	94.65	92.54	86.23	84.73	96.76	94.34	86.51	84.18
TFCCL	Accuracy	99.10	98.95	93.88	93.45	98.84	98.66	91.19	89.02	98.91	98.78	93.71	91.95	99.03	98.45	95.48	93.98	98.95	98.42	95.20	92.22
	Precision	99.30	99.49	90.98	89.93	98.98	98.83	91.60	92.18	98.55	98.23	94.35	92.24	99.02	98.57	93.14	89.62	99.42	98.06	90.10	87.58
	Recall	99.28	98.85	92.82	92.99	99.04	97.94	87.65	82.10	98.65	98.40	90.16	87.50	99.51	98.00	94.15	94.37	99.01	99.62	96.51	89.53
	F1 Score	99.29	99.17	91.67	91.16	99.01	98.38	89.23	86.09	99.10	98.31	91.53	89.45	99.26	98.27	93.47	91.48	99.21	98.83	93.07	88.37

under various noise ratios. To be specific, the effects of the pre-training, the FCCL strategy, the UMAP and the dynamic UFCM updating strategy in TFCCL are verified. The results of the ablation study are illustrated in Table VIII. Based on the results, detailed analyses are given as follows.

- 1) The pretraining helps the Transformer encoder exploit the internal structure and capture useful feature representations. When trained directly for classification without pre-training, the average performance of the TFCCL framework declines with increasing noise ratios.
- 2) The FCCL strategy significantly improves the performance of outlier detection with noisy labels by utilizing the intrinsic features of data. Employing the UMAP is able to enhance the performance of the FCCL strategy when dealing with severe label noise.

- 3) The dynamic UFCM updating strategy effectively stabilizes the training process and the final performance.
- 4) Each of the main components plays a critical role in the TFCCL framework. The best performance of TFCCL can be achieved only if all components work together.

V. CONCLUSION

In this article, a new TFCCL framework has been developed for LNL in outlier detection on industrial time series. Specifically, a novel FCCL strategy has been introduced to update the Transformer encoder, where a UFCM algorithm has been proposed for identifying positive and negative sample pairs. A dynamic two-stage training scheme has been designed to train the outlier detector, incorporating a joint learning strategy to enhance its robustness against noisy labels. Moreover, a

dynamic updating strategy has been employed to update the UFCM algorithm throughout the training process. To further improve model reliability, a label-consistency regularization term has been introduced to minimize the discrepancy between the outputs of the outlier detector and the UFCM algorithm. Experimental results have demonstrated the effectiveness of the developed TFCCL framework. Future research directions can be summarized as follows.

- 1) Utilizing optimization techniques for automatic hyperparameter selection [7], [16], [28], [34], [35].
- 2) Extending the developed TFCCL framework to multiclass outlier detection tasks [21], [22].
- 3) Modifying the Transformer encoder architecture to enhance feature extraction performance [3], [27].
- 4) Refining the FCCL strategy by adjusting the contrastive loss and improving the clustering algorithm [24], [27].

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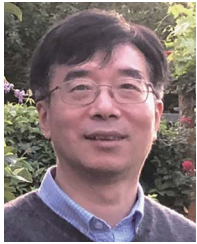


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